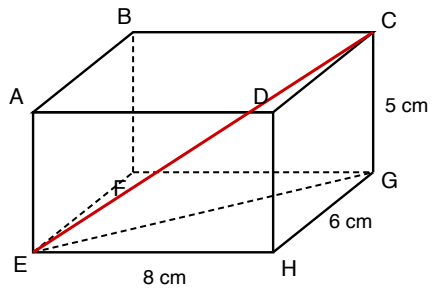
Shape and Space 9 — 3D trigonometry

Calculator allowed. Marks in brackets. ★ = harder question. Give lengths to 3 s.f. and angles to 1 d.p. unless told otherwise. Diagrams are not drawn accurately.

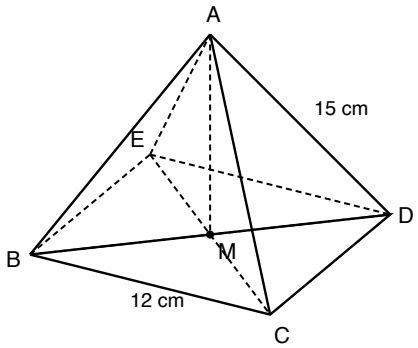
1 ABCDEFGH is a cuboid with EH $= 8$ cm, HG $= 6$ cm and CG $= 5$ cm. Find **a** the length EG **b** the angle between the diagonal EC and the base EFGH. [6]



a Triangle EHG is right-angled at H: $\mathrm{EG} = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ cm.

b The angle between EC and the base is angle CEG, in the right-angled triangle CEG (CG is vertical): $\tan\theta = \dfrac{\mathrm{CG}}{\mathrm{EG}} = \dfrac{5}{10} \Rightarrow \theta = 26.6°$.

2 ABCDE is a square-based pyramid. The base BCDE is horizontal with side 12 cm, and AB $=$ AC $=$ AD $=$ AE $= 15$ cm. M is the centre of the base, so AM is vertical. [13]



a Calculate **i** BD **ii** BM **iii** AM. **b** Calculate the angle that AD makes with the base.

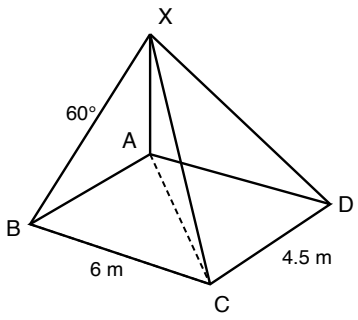
c Calculate the angle between AM and the face ABC.

a i BD is the diagonal of the square base: $\sqrt{12^2 + 12^2} = 12\sqrt{2} = 16.97$ cm. **ii** $\mathrm{BM} = \frac{1}{2}\mathrm{BD} = 8.485$ cm. **iii** Triangle ABM is right-angled at M: $\mathrm{AM} = \sqrt{15^2 - 8.485^2} = \sqrt{153} = 12.37$ cm.

b Angle ADM, with DM $=$ BM: $\cos\theta = \dfrac{\mathrm{DM}}{\mathrm{AD}} = \dfrac{8.485}{15} \Rightarrow \theta = 55.6°$.

c Let N be the mid-point of BC; AN lies in the face ABC and MN $= 6$ cm (half the side). The angle between AM and the face is angle MAN in the right-angled triangle AMN: $\tan\phi = \dfrac{\mathrm{MN}}{\mathrm{AM}} = \dfrac{6}{12.37} \Rightarrow \phi = 25.9°$.

3 The diagram shows the roof of a kiosk. ABCD is a horizontal rectangle with AB $= 6$ m and BC $= 4.5$ m. The point X is vertically above A, and the edge XB makes an angle of 60° with the horizontal. Calculate the angle that the ridge XC makes with the horizontal. [6]



First the height: in the vertical right-angled triangle XAB,
$\mathrm{XA} = \mathrm{AB}\tan 60° = 6\tan 60° = 10.39$ m.
Then the horizontal distance from A to C, the diagonal of the rectangle:
$\mathrm{AC} = \sqrt{6^2 + 4.5^2} = 7.5$ m.
Finally in the vertical triangle XAC: $\tan\theta = \dfrac{\mathrm{XA}}{\mathrm{AC}} = \dfrac{10.39}{7.5} \Rightarrow \theta = 54.2°$.

Total: 25 marks

Check: angle between a line and a plane: drop a perpendicular from the top of the line to the plane, join the foot to where the line meets the plane, use that right-angled triangle · redraw each 2D triangle with its lengths · the base diagonal of a cuboid/square needs Pythagoras first · keep full calculator values between steps.